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Modeling, Control, and Closed-Loop Mobility Characterization of a Spherical Sailing Omnidirectional Rover (SSailOR)

This paper presents a control-oriented dynamic model, controller, and closed-loop mobility characterization for the first wind-powered spherical rover capable of net upwind motion. This device, termed the Spherical Sailing Omnidirectional Rover (SSailOR), incorporates design features within a spherical, terrestrial rover that mimic the role that a centerboard (or keel) and lifting sails play in allowing net upwind motion for sailboats. Specifically, a traction hoop enables significant lateral resistance, thereby providing a nonholonomic constraint in the direction of travel. Lifting sails enables net thrust even when traveling significantly upwind, while also providing heading control. While providing unique capabilities, the SSailOR gives rise to a complex design and control space, where careful model-based design and control are necessary to ensure that the SSailOR can simultaneously (i) make net upwind progress, (ii) respond quickly to wind speed/direction changes, (iii) limit heel angle, and (iv) control its heading. To simultaneously address these challenges, we first present a control-oriented dynamic model. This is followed by the presentation of a combined heading and heel angle controller. Finally, with the dynamic model and control structure in place, we present a detailed closed-loop Pareto analysis, which illustrates the tradeoff between transient and steady-state performance, along with the design features that favor one modality of performance over another.
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Keywords: autonomous systems, robotics, energy systems, and control

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1 Introduction

There are major challenges to scientific data collection in remote and hostile regions, such as the freezing arctic tundra or the arid sands of the Sahara Desert. Lack of infrastructure in remote regions makes transport logistics expensive, and inclement weather commonly causes delays. Arctic research is particularly challenging, as manned data collection is only possible in the summer, given that the region is shrouded in darkness and bitter cold for the remainder of the year [1].

Unmanned rovers have emerged in recent decades as a solution to the challenges in these regions. Harnessing environmental energy sources, wind and solar-powered rovers have been proposed for long-term and persistent exploration missions that allow data collection without human reliance, thus lessening the burden and cost of research in these remote regions. Wheeled solar-powered rovers such as *Solero* [2] and the *Cool Robot* [3] were developed for autonomous roving missions. However, because the polar regions do not receive light for half the year, these rovers are incapable of running persistent missions in arctic environments and thus are not suitable for unmanned persistent data collection missions. Beyond the lack of a reliable solar resource that renders wheeled solar vehicles unsuitable, the high wind speeds common to polar [4] and desert [5] regions present another issue for both wind- and solar-powered wheeled rovers: tipping. This issue, which wheeled land sailer designs especially suffer from due to large lateral sail forces, can immediately end a mission or require human intervention, thus rendering wheeled rovers unfit to carry out unmanned persistent missions reliably.

To alleviate the issue of tipping, some recent efforts have focused on wind-powered spherical rover designs that extract energy from the abundant wind resource, and which are incapable of tipping due to the spherical geometry (e.g., Ref. [6]). This concept has been studied in both Martian [7] and Antarctic (Refs. [6] and [8]) applications. While existing spherical, wind-powered rovers overcome the issues of tipping and reliance on an inconsistent solar resource, these designs share a common limitation. Specifically, existing spherical rovers are drag-based, and are thus incapable of making upwind progress (in fact, some lack *any* directional control authority). Consequently, in mission domains with a consistent prevailing wind direction, persistent missions become impossible. Furthermore, lack of net omnidirectional mobility makes it often impossible for such rovers to target specific areas of identified interest for the mere reason that those locations happen to be upwind. As such, tumbleweed rovers are usually sent on one-off missions until they get stuck or reach a predefined end point in the mission domain. The combination of one-time use and inability to take a deterministic path prohibits tumbleweed rovers from succeeding in meeting mission goals.

This work details the development of a dual-sail, wind-powered spherical rover capable of making net upwind progress, thereby enabling persistent operation without the threat of tipping and requiring human intervention. Termed the Spherical Sailing Omnidirectional Rover (SSailOR) and depicted in Fig. 1, this rover overcomes limitations of the aforementioned wind-powered rovers—land-sailers being unable to recover from tipping and drag-based tumbleweed rovers being incapable of upwind motion. As shown in Table 1, The SSailOR has the positive traits of other wind-powered rovers, while overcoming their limits. The SSailOR accomplishes this through several features:

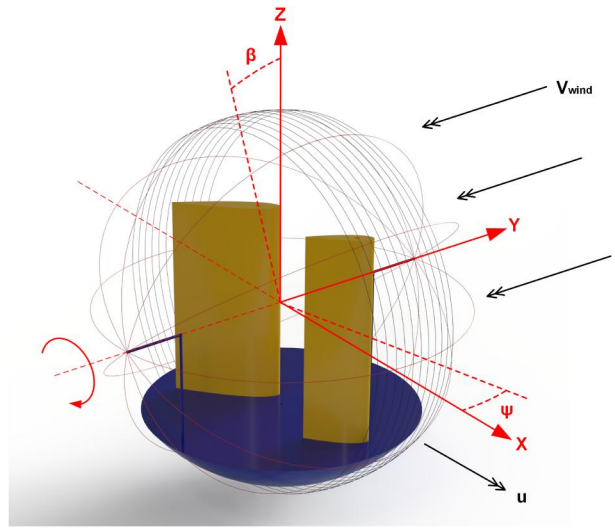

Fig. 1 CAD rendering of a full-scale SSailOR with a 1 m diameter. This rendering depicts SSailOR's speed (u), heading angle (ψ), and heel angle (β), along with the wind velocity (V_{wind}). The traction hoop (black) is rubberized for grip.

Table 1 Major distinctions between other wind-powered rovers and the SSailOR

	Self-righting	Upwind-capable
Land sailer	×	√
Tumbleweed rover	√	×
SSailOR	√	√

- Two airfoil-based sails provide high lift-to-drag ratios at limited angles of attack, thereby resulting in net forward thrust even when the apparent wind direction (referring to the wind "felt" by the moving rover) is as little as 40 degrees off of the rover's heading.
- A ballasted hull (which holds the on-board electronics and to which the sails are attached) is suspended in the sphere and provides a stabilizing roll moment when lateral forces on the sail cause the assembly to heel.
- A traction hoop remains in contact with the ground at all times that the heel angle is maintained below 15 deg (see Figs. 2 and 3), providing a lateral reaction force (like a tire) that constrains the SSailOR to translate in the direction that the hoop is rolling.
- The differential action of the two sails (feathering one while trimming the other) provides directional control.

In this paper, we first present a control-oriented dynamic model of the SSailOR, which incorporates characterizations of its translational, yaw, and heel dynamics, which are driven by detailed aerodynamic models for the sails while also accounting for rolling resistance. The aerodynamic coefficients in this model have been informed by scaled wind tunnel testing, which is also discussed in the paper. Following the presentation of the model, we present a multivariable controller that simultaneously manages three objectives: (i) maximizing net longitudinal force, (ii) tracking a target heading, and (iii) maintaining heel angle at or below a prescribed threshold. It should be noted that these objectives can be in conflict, as (for example), a controller that maximizes net longitudinal force will often result in a yaw moment that must be zeroed at some sacrifice to performance. These tradeoffs, and the impact of the design on the extent of these tradeoffs, are discussed in more detail in the remainder of the paper. We present simulation results that demonstrate the rover's ability to track a straight line, tack upwind, and respond to gusts. Finally, we examine the tradeoff between

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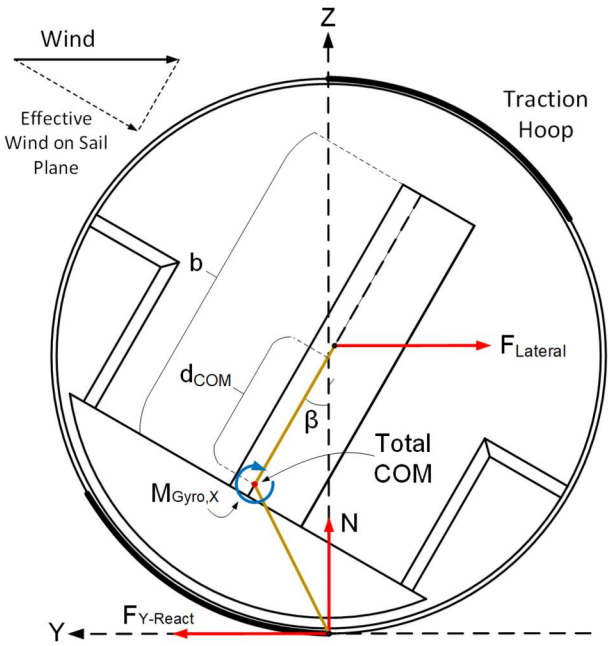

Fig. 2 Moment balance diagram for heel angular acceleration. All moments are about the total center of mass. The traction hoop must maintain ground contact to satisfy the no-slip condition.

steady-state performance (represented through the SSailOR's velocity polar plot) and transient performance, along with the effect of critical design parameters on the extent of this tradeoff.

The remainder of this work is structured as follows: Section 2 details the design and corresponding dynamic model for the SSailOR. Section 3 details the proposed control architecture. Section 4 summarizes the wind tunnel setup and testing done to characterize the lift and drag of the sails as well as the drag force of the rover's cage and hull. Section 5 presents dynamic simulation results using the proposed control architecture, demonstrating both the closed-loop tracking performance of the proposed control architecture and the impact of critical design parameters on real-time performance. Finally, conclusions drawn from this work are discussed in Sec. 6.

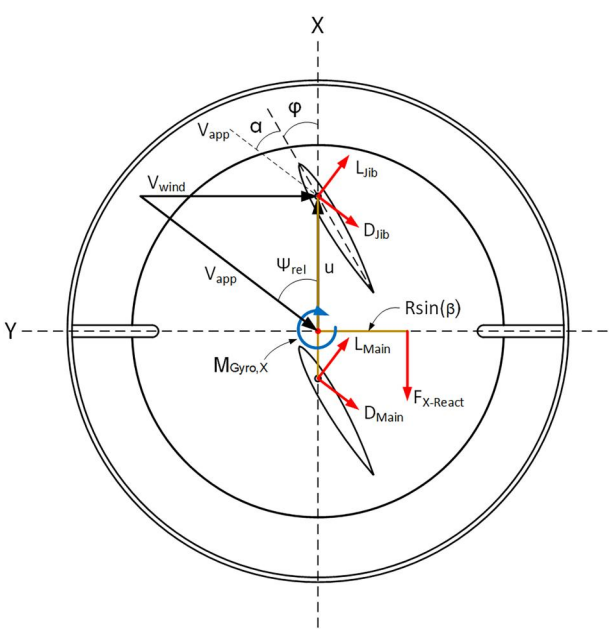

Fig. 3 Top-down diagram of apparent wind and forces on the rover

2 Control-Oriented Dynamic Model

2.1 Rover Design. As shown in Fig. 1, the SSailOR features two rigid wing-sails—a jib (front) sail and a main (aft) sail—mounted on a hull within a rotating spherical cage. The spherical cage uses a "traction hoop" to increase the area of ground contact, thereby forcing the rover to roll in the direction of the traction hoop without lateral slip. Note that this hoop does not entirely enclose the rover, which avoids obstructing incoming wind from reaching the sails. However, the implication of this is that the SSailOR must maintain a sufficiently low heel angle to keep the traction hoop in contact with the ground. Here, heel angle (β) is the roll angle about the SSailOR's longitudinal X-axis (as depicted in Fig. 1). The hull inside the spherical cage is constrained such that the hull's longitudinal axis is always in line with the traction hoop, but without rolling with the traction hoop, not unlike an inverted hamster wheel. The hull also provides the rover with a low center of gravity, enabling it to self-right itself against lateral forces on the sails and rendering the SSailOR incapable of tipping over.

2.2 Dynamic Modeling. This subsection details the development of a dynamic model of the SSailOR. This model is composed of seven state variables contained within the state vector $\mathbf{X}$:

$$\mathbf{X} = [x \quad y \quad u \quad \psi \quad r \quad \beta \quad h]^{\mathrm{T}} \tag{1}$$

where definitions for each state variable are given in Table 2. In addition to being influenced by the wind speed and direction (V_{wind} and ψ_{wind}, respectively), the evolution of the state variables is influenced by two control variables: the main sail angle, φ_{main}, and the jib sail angle, φ_{jib}. Wind speed and wind angle are external factors in the system.

Several key assumptions are made in deriving the equations of motion, which are justified by features of the design and ultimately simplify the equations by removing Coriolis and gyroscopic coupling terms that are found in more general 3D equations of motion:

- By its design, the traction hoop results in a no-sideslip kinematic constraint when in contact with the ground. To understand the accuracy of this assumption, given its rubber exterior and the number of "rings" contained in the traction hoop, its cornering stiffness [9] is estimated to be approximately 15 Newtons/deg for a normal force of 49 Newtons (consistent with the weight of the nominal design). Under maximum anticipated aerodynamic forces, this will result in no more than a one-degree sideslip angle. If the heel angle exceeds 15 degrees, the traction hoop will lose contact with the ground, and the no-sideslip constraint will become invalid (which is the reason why a heel angle controller has been introduced in this work). Thus, the only translational motion is in the body-x direction. Note that this neglects body-y and body-z translational velocities that result from heel rate (h); however, these velocities are consistently at least two orders of magnitude lower than the translational velocities in the x direction.

Table 2 Definitions and units for the seven state variables

Variable	Definition	Units
x	Horizontal position in ground frame	m
y	Vertical position in ground frame	m
u	Velocity in the rover's longitudinal direction	$\mathrm{m \cdot s^{-1}}$
ψ	Heading of the rover	rad
r	Yaw rate	$\mathrm{rad \cdot s^{-1}}$
β	Heel angle of the rover (the angle at which it is tipping to its side, see 2)	rad
h	Rate of change of the heel angle	$\mathrm{rad \cdot s^{-1}}$

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- The cage exhibits radial symmetry, and the hull exhibits symmetry about both the $x-z$ and $y-z$ planes, thereby rendering the products of inertia (i.e., I_{xy}, I_{yz}, I_{xz}) equal to zero.
- The angular velocities of the cage about its x and z axes are small (in general, at most two orders of magnitude smaller in all simulations) compared to the rotation of the cage about its y axis. Due to the fact that the x and z angular velocities are inconsequential even when at their greatest values (i.e., during tacking maneuvers where heading and heel angle change quickly), these angular velocities are not included in velocity, acceleration, and time derivative of angular momentum expressions.

Under the aforementioned assumptions, the state equations are as follows:

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{u} \\ \dot{\psi} \\ \dot{r} \\ \dot{\beta} \\ \dot{h} \end{bmatrix} = \begin{bmatrix} u\cos\psi \\ u\sin\psi \\ \frac{1}{m_{\text{eq}}}\sum F_{\text{longitudinal}} \\ \frac{r}{\cos\beta} \\ \frac{1}{J_{\text{zz}}}\sum M_{\text{zz}} \\ h \\ \frac{1}{J_{\text{xx}}}\sum M_{\text{xx}} \end{bmatrix} \tag{2}$$

where definitions and units for M_{xx}, M_{zz}, and $F_{\text{longitudinal}}$ are given in Table 3. Note that the moment terms, M_{xx} and M_{zz}, include contributions from aerodynamic forces, ground reaction forces, and gyroscopic coupling terms, as spelled out subsequently. Rather than separately characterizing the longitudinal rotational dynamics of the cage (whose rotational velocity is kinematically constrained with respect to its translational velocity under the no-slip condition), an equivalent mass model is used to account for the rotational acceleration of the spherical cage (where the system's equivalent mass is the sum of the hull/sail mass, cage mass, and reflected mass of the cage). Based on a zero longitudinal slip assumption, the cage's equivalent mass $m_{\text{cage,eq}}$—the sum of its mass and reflected mass (the effective additional translational mass arising from the body's rotational inertia)—is derived from Ref. [10] and acquired using the following equation:

$$m_{\text{cage,eq}} = m_{\text{cage}} + \frac{J_{\text{cage,P}}}{R^2} \tag{3}$$

where the spherical cage is assumed to be a thin-walled sphere of radius R for inertial calculations, and $J_{\text{cage,P}}$ is the moment of inertia of the cage about its principal axis (center of the sphere). The hull of the rover is modeled as a homogeneous spherical cap of height H and radius R rotating about the total center of mass of the system.

Excluding the mass and inertia of the sails, the equivalent mass and inertia (about the total center of mass) for the entire system are given as follows:

$$\begin{aligned} m_{\text{eq}} &= m_{\text{hull}} + m_{\text{cage,eq}} \\ J_{\text{zz}} &= J_{\text{zz,hull}} + J_{\text{zz,cage}} \\ J_{\text{xx}} &= J_{\text{xx,hull}} + J_{\text{xx,cage}} \end{aligned} \tag{4}$$

Table 3 Definitions and units for state equation moments and forces

Variable	Definition	Units
M_{xx}	Roll moment about the total center of mass in the body frame	N · m
M_{zz}	Yaw moment about the total center of mass in the body frame	N · m
$F_{\text{longitudinal}}$	Sum of the x components of the aerodynamic forces minus the rolling resistance	N

In this dynamic model, the forces and moments depend on what is known as the *apparent wind velocity*, which is the velocity of the wind relative to the moving SSailOR. Note that, for simplicity of modeling, a uniform wind profile (which neglects boundary layer effects near the ground) is considered throughout this work. Prior to calculating the apparent wind velocity, the model first computes a projection of the body-frame wind velocity in the direction perpendicular to the sail plane, accounting for heel angle. Assuming the wind is two-dimensional in the X–Y plane (a common practice in sailing literature—see Refs. [11,12], and [13]—which can be applied to flat terrain like tundra and desert, where wind is not broken by obstacles), the effective wind speed vector $\mathbf{V}_\perp$ can be calculated as shown in Ref. [14]:

$$\begin{aligned} \mathbf{V}_\perp &= \mathbf{V}_{\text{wind}} - (\mathbf{V}_{\text{wind}} \cdot \hat{\mathbf{m}})\hat{\mathbf{m}} \\ \hat{\mathbf{m}} &= \begin{bmatrix} 0 \\ -\sin\beta \\ \cos\beta \end{bmatrix} \\ \mathbf{V}_{\text{rel}} &= -V_{\text{wind}} \begin{bmatrix} \cos\psi_{\text{rel}} \\ \sin\psi_{\text{rel}} \\ 0 \end{bmatrix} \end{aligned} \tag{5}$$

where $\psi_{\text{rel}} = \psi_{\text{wind}} - \psi$. Because the wind is assumed to be X–Y planar, the z-component of $\mathbf{V}_\perp$ is disregarded. Next, the apparent wind vector ($\mathbf{V}_{\text{app}}$) is obtained by subtracting the longitudinal velocity (u) from the effective wind vector ($\mathbf{V}_\perp$):

$$\mathbf{V}_{\text{app}} = \begin{bmatrix} V_{\perp x} \\ V_{\perp y} \\ 0 \end{bmatrix} - \begin{bmatrix} u \\ 0 \\ 0 \end{bmatrix} \tag{6}$$

The magnitude (V_{app}), angle with the longitudinal axis (ψ_{app}), and unit vector ($\hat{\mathbf{V}}_{\text{app}}$) of the apparent wind are necessary for subsequent force and moment calculations.

All of the forces and moments on the sails (which contribute to $F_{\text{longitudinal}}, M_{xx}$, and M_{zz}) depend directly on the control signals (φ_{jib}; φ_{main}) through their contribution to the sail's *angle of attack*. The angle of attack (α) for each sail is determined by $\alpha_i = \psi_{\text{app}} - \varphi_i$, where $i \in \{\text{jib}, \text{main}\}$. The present work models the sails in a nonstalled configuration, with future work aimed at characterizing the stalled configuration as well (which is especially relevant for direct-downwind and nearly downwind operation). In practice, headings that are off by at least 30 degrees from a direct downwind heading (where the exact number depends on wind speed) have been shown not to lead to stall. Under the nonstalled assumption, the lift coefficient (C_L) and magnitudes for lift force $|F_{L_i}|$ and drag force $|F_{D_i}|$ are estimated as follows (as was done for a robotic sailboat in Ref. [11]):

$$\begin{aligned} |F_{L_i}| &= \frac{1}{2}\rho|\mathbf{V}_{\text{app}}|^2 C_{L_i} S_i, \quad C_{L_i} = k_{1_i}\alpha_i \\ |F_{D_i}| &= \frac{1}{2}\rho|\mathbf{V}_{\text{app}}|^2 \left(C_{D,0_i} + \frac{C_{L_i}^2}{AR_i}\pi e\right) S_i \\ & i \in \{\text{jib}, \text{main}\} \end{aligned} \tag{7}$$

where the definitions and units for variables in the lift and drag equations are given in Table 4. Drag force acts parallel to the apparent wind, whereas lift acts perpendicular to it. Using the unit vector $\hat{\mathbf{V}}_{\text{app}}$, the drag force vector is simply

$$\mathbf{F}_{D_i} = |F_{D_i}|\hat{\mathbf{V}}_{\text{app}} \tag{8}$$

To account for both the symmetry over the sail's longitudinal axis and the possibility of a negative angle of attack, the lift force vector is calculated as

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Table 4 Definitions and units for lift and drag equations

Variable	Definition	Units
k_{1_i}	The lift curve slope of each wing sail	rad^{-1}
$\boldsymbol{\rho}$	The density of air	$\text{kg} \cdot \text{m}^{-3}$
S_i	The planform area of each sail	m^2
e	The sail's Oswald efficiency factor	—
$C_{\text{D},0_i}$	The zero-lift drag coefficient of each sail	—

$$\mathbf{F}_{L_i} = |F_{L_i}|\text{sign}(\alpha)\text{sign}(\psi_{\text{app}})\hat{\mathbf{V}}_{\text{app}\perp}$$
$$\hat{\mathbf{V}}_{\text{app}\perp} = \begin{bmatrix} -\hat{V}_{\text{app},y} \\ \hat{V}_{\text{app},x} \\ 0 \end{bmatrix} \tag{9}$$

When the wind is coming from the right side of the rover (starboard tack), the angle of attack and apparent wind angle are both negative. Including the signs of each ensures that the lift acts in the proper direction with respect to the angle of attack and apparent wind angle.

To approximate the windage drag caused by the hull, spherical cage, traction hoop, and supports—which connect the hull to the spherical cage as a revolute joint—the following equation is used:

$$\begin{aligned} \mathbf{F}_{D_{\text{windage}}} &= k_{\text{windage}}|\mathbf{V}_{\text{app}}|^2\hat{\mathbf{V}}_{\text{app}} \\ k_{\text{windage}} &= \frac{1}{2}\rho\sum A_i C_{D_i} \\ i &\in \{\text{hull, cage, hoop, support}\} \\ A_{\text{cage/hoop}} &= 2Rn\tau \\ A_{\text{support}} &= 2\tau_{\text{support}} \end{aligned} \tag{10}$$

The spherical cage and traction hoop are comprised of n tori ("rings"), and the major (hoop diameter $2R$) and minor (hoop thickness τ) diameters were used to approximate windage drag based on the experimental results and the value of $C_{D_{\text{hoop}}}$ from Ref. [15]. The supports are simple cylinders of thickness τ_{support}, and used the literature drag of a cylinder from Ref. [16]. The windage drag of the hull was found via wind tunnel experimentation, as detailed in Sec. 4.

Finally, the force due to rolling resistance is given by

$$F_{\text{rr}} = mgC_{\text{rr}} \tag{11}$$

With expressions for the lift, drag, and rolling resistance in place, longitudinal acceleration is obtained as follows, where the subscript x refers to the x-component of a given force:

$$\dot{u} = \frac{F_{\text{L}_{\text{jib}},x} + F_{\text{D}_{\text{jib}},x} + F_{\text{L}_{\text{main}},x} + F_{\text{D}_{\text{main}},x} - F_{\text{rr}} + F_{D_{\text{windage}},x}}{m_{\text{eq}}} \tag{12}$$

Next, the lateral components of the forces are used to find the angular acceleration of the heel ($\dot{h}$). The moments M_N, M_{Lateral}, $M_{Y-\text{React}}$, and $M_{\text{Gyro},X}$ are produced by the normal force, the lateral forces from the sails, the ground reaction force in the lateral direction, and the gyroscopic moment about the X-axis, respectively, which are illustrated in Fig. 2. It is important to note that the reaction force maintains the no-slip condition and includes centripetal forces experienced when turning. Given these moments and the rotational inertia of the rover, the angular acceleration of the heel is calculated using the following moments about the body-x axis:

$$\begin{aligned} M_{N,x} &= -d_{\text{COM}}mg\sin\beta \\ M_{\text{Lateral},x} &= -\left(d_{\text{COM}} + \frac{b}{2} - R + H\right)\cos\beta\sum\left(F_{L_{i,y}} + F_{D_{i,y}}\right) \\ M_{Y-\text{React},x} &= (R - d_{\text{COM}}\cos\beta)\left(\text{mur} - \sum\left(F_{L_{i,y}} + F_{D_{i,y}}\right)\right) \\ M_{\text{Gyro},X} &= \frac{J_{\text{yy}_{\text{cage},P}}}{R}\text{ur} \\ i &\in \{\text{jib, main}\} \end{aligned} \tag{13}$$

Table 5 Definitions and units for heel moments

Variable	Definition	Units
b	The span of the wing-sail	m
d_{COM}	The distance from the sphere's center to the combined center of mass of the hull and cage	m
$J_{\text{yy}_{\text{cage,P}}}$	The moment of inertia of the cage about the principle (center) Y-axis	$\text{kg} \cdot \text{m}^2$
m	True total mass of the SSailOR	kg

where the definitions for b, d_{COM}, $J_{\text{yy}_{\text{cage},P}}$, and m are given in Table 5. Note that, in the equation for $M_{Y-\text{React},x}$, the *mur* term represents the centrifugal force experienced by the SSailOR when adjusting heading. All aerodynamic forces generated by the wing-sails are assumed to operate at quarter-chord and midspan of each sail. Using these expressions for the moments about the rover's body-x axis (running through the total COM), the angular acceleration of the heel angle is calculated as

$$\dot{h} = \frac{M_{N,x} + M_{\text{Lateral},x} + M_{Y-\text{React},x} + M_{\text{Gyro},X}}{J_{\text{xx}}} \tag{14}$$

The last calculation needed to model the dynamics of the SSailOR is the moment balance of the sail forces about the body z-axis of the sphere, which is the axis of symmetry through the sphere and hull, as seen in Fig. 3. The moment arm of each sail ($\mathbf{r}_{\text{qc}_{\text{jib}}}$ and $\mathbf{r}_{\text{qc}_{\text{main}}}$) is the distance from the axis of symmetry to the quarter chord of each sail and is given by

$$\begin{aligned} \mathbf{r}_{\text{qc}_{\text{jib}}} &= \begin{bmatrix} \ell_{\text{jib}} \\ 0 \\ 0 \end{bmatrix} - d_{\text{qc}_{\text{jib}}}\begin{bmatrix} -\cos\varphi_{\text{jib}} \\ -\sin\varphi_{\text{jib}} \\ 0 \end{bmatrix} \\ \mathbf{r}_{\text{qc}_{\text{main}}} &= \begin{bmatrix} -\ell_{\text{main}} \\ 0 \\ 0 \end{bmatrix} - d_{\text{qc}_{\text{main}}}\begin{bmatrix} -\cos\varphi_{\text{main}} \\ -\sin\varphi_{\text{main}} \\ 0 \end{bmatrix} \end{aligned} \tag{15}$$

where ℓ_{jib} and ℓ_{main} are the distances from the rover's center to the mast of each sail, and $d_{\text{qc}_{\text{jib}}}$ and $d_{\text{qc}_{\text{jib}}}$ are the distances from the mast to the quarter-chord of each sail. The six moments acting on the rover are calculated as

$$\begin{aligned} \mathbf{M}_{L_i} &= \mathbf{r}_{qc_i} \times \mathbf{F}_{L_i} \\ \mathbf{M}_{D_i} &= \mathbf{r}_{qc_i} \times \mathbf{F}_{D_i} \\ i &\in \{\text{jib, main}\} \\ M_{X-\text{React},z} &= R\sin\beta\left(F_{\text{rr}} + \frac{J_{\text{cage},P}}{R^2}\dot{u}\right) \\ M_{\text{Gyro},z} &= -\frac{J_{\text{yy}_{\text{cage},P}}}{R}uh \end{aligned} \tag{16}$$

Using these moments, the yaw acceleration (time derivative of yaw rate) is given by

$$\dot{r} = \frac{M_{L,z} + M_{D,z} + M_{X-\text{React},z} + M_{\text{Gyro},z}}{J_{\text{zz}}} \tag{17}$$

where

$$\begin{aligned} M_{L,z} &= M_{L_{\text{jib}},z} + M_{L_{\text{main}},z} \\ M_{D,z} &= M_{D_{\text{jib}},z} + M_{D_{\text{main}},z} \end{aligned}$$

3 Methods of Control

This section details the development of a real-time control architecture for maximizing the SSailOR's speed while ensuring that ground contact with the traction hoop is maintained and a

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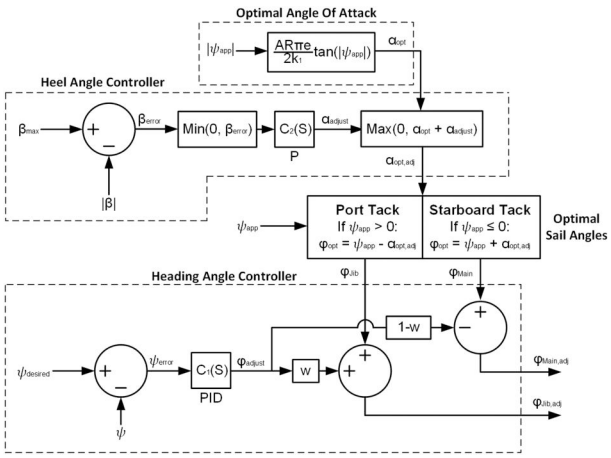

Fig. 4 Block diagram of the SSailOR controller

desired heading is tracked. To accomplish these objectives, a parallel control architecture is proposed, as illustrated in Fig. 4. This controller is composed of four primary components:

(1) A subsystem wherein the optimal angle of attack is calculated. This is the angle of attack that maximizes the net forward (longitudinal) force generated by the sails at any given apparent wind angle (ψ_{app}).
(2) A controller to limit heel angle such that ground contact with the traction hoop is maintained.
(3) A switching gain block that modifies the sign on the desired angle of attack if the apparent wind direction changes.
(4) A controller to track a desired heading by modifying the net yaw moment generated by the sails.

These four components combine to determine the sail angles for the jib and main sail. In this work, it is assumed that the desired sail angles can be achieved instantaneously, as was done in Ref. [11].

3.1 Optimal Angle of Attack. The optimal angle of attack, α_{opt}, is the angle of attack that maximizes the net forward force in the rover's longitudinal direction. The directions of the lift and drag forces are related to the apparent wind angle ψ_{app}, which will decrease as the rover accelerates (for a given wind direction and rover heading). Due to this directional dependence on apparent wind angle, a measurement of ψ_{app} is used to calculate α_{opt}.

The magnitude of the net forward force for a particular sail in the longitudinal direction is calculated as

$$F_{\text{net}_i} = \frac{\rho|\mathbf{V}_{\text{app}}|^2 S_i}{2}\left(k_{1_i}\alpha_i \sin\psi_{\text{app}} - \left(C_{D,0} + \frac{{k_{1_i}}^2{\alpha_i}^2}{AR_i\pi e}\right)\cos\psi_{\text{app}}\right) \tag{18}$$

To calculate α_{opt_i}, the partial derivative of F_{net_i} with respect to α, is taken and set equal to zero, resulting in the following algebraic equation for α_{opt_i}:

$$\alpha_{\text{opt}_i} = \frac{AR_i\pi e}{2k_{1_i}}\tan\psi_{\text{app}} \tag{19}$$

The optimal angle of attack is clamped between a maximum and a minimum value $-\alpha_{\max} \leq \alpha_{\text{opt}} \leq \alpha_{\max}$. Limiting the maximum and minimum allowable angle of attack ensures that each sail will not stall.

3.2 Heel Controller. The heel controller is necessary to keep the traction hoop of the SSailOR in contact with the ground and prevent the rover from slipping. Keeping the heel angle low also preserves a higher effective wind speed on the sails, as demonstrated in Eq. (5). When the magnitude of the heel angle exceeds the threshold $\beta_{\max}$, the heel controller curtails the angle of attack setpoint. This controller takes in β, $\beta_{\max}$ (the maximum allowable heel angle), and α_{opt}, outputting the adjusted optimal angle of attack $\alpha_{\text{opt,adj}}$, which maximizes net longitudinal force under the constraint of a maximum heel angle. As displayed in the "Limiting Heel Angle" section of Fig. 4, the absolute value of heel angle is subtracted from the maximum allowable heel angle threshold to obtain a heel error, β_{error}, which will be positive when β is below $\beta_{\max}$, but negative when above it. To ensure that the heel angle controller does not attempt to increase the heel angle when it falls below this maximum allowable value, the minimum between 0 and β_{error} is taken. The minimum value is then fed into a wind speed-scaled proportional controller given by

$$C_2(s) = G_2 V_{\text{wind}} \tag{20}$$

where G_2 is a constant gain. Wind speed-based scaling is required since higher wind speeds increase the forces generated by the sails, thereby requiring a higher gain to curtail those forces and limit heel. The output of $C_2(s)$ is a negative angle of attack adjustment, α_{adjust}. This term is added to α_{opt} to calculate $\alpha_{\text{opt,adj}}$.

3.3 Optimal Sail Angles and Limits. Given that the sails are the only actuators in the system, the only physical input to the system is a pair of sail angles. To calculate the desired sail angles (prior to modifying these angles as required by the heading controller), only the apparent wind angle and the adjusted optimal angle of attack are needed. Because $\alpha_{\text{opt,adj}}$ is always positive, piecewise calculation of the sail angles is performed depending on whether the wind is coming from the port (left) or starboard (right) side of the rover. The piecewise function used in the "Calculating Optimal Sail Angles" section of Fig. 4 is

$$\varphi_{\text{jib}}, \varphi_{\text{main}} = \begin{cases} \psi_{\text{app}} - \alpha_{\text{opt,adj}}, & \text{if} \quad \psi_{\text{app}} > 0 \\ \psi_{\text{app}} + \alpha_{\text{opt,adj}}, & \text{if} \quad \psi_{\text{app}} \leq 0 \end{cases} \tag{21}$$

These resulting sail angles are those needed for each sail to operate at the adjusted optimal angle of attack for the current wind condition.

Finally, the sail angles are subject to the following saturation limits:

$$\begin{gathered} -\frac{\pi}{2} \leq \varphi_{\text{jib,adj}}, \quad \varphi_{\text{main}} \leq \frac{\pi}{2} \\ \begin{cases} -\theta \leq \varphi_{\text{jib,adj}}, \quad \varphi_{\text{main}} \leq \psi_{\text{app}}, & \text{if} \quad \psi_{\text{app}} > 0 \\ \psi_{\text{app}} \leq \varphi_{\text{jib,adj}}, \quad \varphi_{\text{main}} \leq \theta, & \text{if} \quad \psi_{\text{app}} \leq 0 \end{cases} \end{gathered} \tag{22}$$

The first constraint is both geometric and aerodynamic. Because the mast is assumed to be at quarter-chord, moving the sail angle past $\pi/2$ could result in contact between the sail and the spherical cage. Additionally, restricting this range ensures that a negative angle of attack is never generated. The second constraint is conditional based on which side of the craft the wind is coming from. This constraint ensures the sail angle never exceeds the apparent wind angle (which will generate lift in the wrong direction). The θ term in this constraint is an angular limit on the sail angles such that if the jib sail has to back-wind (which occurs when ψ_{app} is small and the jib sail operates with drag and lift pointing backward due to the optimal angle of attack and correction) to balance moments, it cannot over-correct past a certain point. This prevents large, rapid changes in sail angle that may otherwise destabilize the system.

3.4 Heading Controller. The heading controller is critical to the SSailOR both when pursuing a straight-line path and when turning. Since the SSailOR does not possess a rudder, heading control requires the sails to be used to generate an appropriate yaw moment, thereby typically requiring some deviation of the main and/or jib sail angles from α_{opt}. The heading controller has four inputs:

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rover heading, the desired heading of the rover (ψ_{desired}), and the jib and main sail angles that correspond to $\alpha_{\text{opt,adj}}$ (φ_{jib} and φ_{main}, respectively). The heading controller ultimately produces a sail angle correction value, which is weighted (based on the extent to which each sail is responsible for generating the required yaw moment) and then applied to the jib and main sail angles to obtain $\varphi_{\text{jib,adj}}$ and $\varphi_{\text{main,adj}}$, as shown in the "Heading Control" section of Fig. 4.

To obtain the adjusted sail angles, the heading error ($\psi_{\text{error}} = \psi_{\text{desired}} - \psi$) is first wrapped to ensure that it stays within $-\pi \leq \psi_{\text{error}} \leq \pi$. This error is then fed into the following PID controller, $C_1(s)$:

$$C_1(s) = K_{\text{P1}} + \frac{K_I}{s} + K_D \frac{Ns}{s+N} \tag{23}$$

where K_{P1}, K_I, K_D, and N are the proportional, integral, and derivative gains and filter coefficient, respectively. A nonscheduled PID controller was deemed the most simple and effective solution to achieve heading control. Based on the results shown in Sec. 5, this controller provides consistent convergence to zero heading error and thus gives confidence in the controller's stability. The output of $C_1(s)$ is a sail angle adjustment, φ_{adjust}, which is added to or subtracted from φ_{jib} and φ_{main} to achieve the required yaw moment. This sail angle adjustment is modified by weighting value w and applied to the sail angles as follows:

$$\begin{aligned} \varphi_{\text{jib,adj}} &= \varphi_{\text{jib}} + w\varphi_{\text{adjust}} \\ \varphi_{\text{main,adj}} &= \varphi_{\text{main}} - (1-w)\varphi_{\text{adjust}} \\ 0 &\leq w \leq 1 \end{aligned} \tag{24}$$

If $w = 1$, then only the jib sail has to deviate from the adjusted optimal angle of attack to balance the yaw moment. In this case, the main sail is free to continue operating at the adjusted optimal angle of attack. All of the subsequent simulation results are in fact based on a value of $w = 1$.

4 Wind Tunnel Aerodynamic Characterization

In order to quantify the aerodynamic parameters related to the sails, hull, traction hoop, and cage, wind tunnel testing was performed on a scale model of the SSailOR. For the sake of brevity, this paper includes a summary of the wind tunnel test setup and resulting component-wise lift and drag characterizations (obtained through least squares parameter identification based on force and moment data). A more detailed record of the setup and all test results is available in Ref. [17]. The details of the setup and results are shown in the following subsections 4.1 and 4.2.

4.1 Wind Tunnel Setup. Testing was performed in the subsonic wind tunnel at North Carolina State University, which features a closed test section measuring 812.8 mm (32 inches) in height and 1143 mm (45 inches) in width. A 40% scale model (400 mm diameter) of the SSailOR prototype was fabricated for these experiments [17] to ensure representative Reynolds numbers at the test velocities, which ranged from approximately 2.0×10^5 to 3.0×10^5.

A blockage correction was performed to account for the presence of the model itself in the test section (and the corresponding increased flow velocity in the vicinity of the model). The blockage ratio was calculated based on projected frontal areas across all tested configurations, and the mathematical correction formula is detailed in Sec. 4.2. This correction follows the approach introduced by Barlow et al. [18,19], and its influence was incorporated into the final parameter identification process. Additionally, the test configuration was designed to maintain a minimum clearance of approximately 20 cm between the outer surface of the SSailOR model and wind tunnel test section walls, further minimizing near-wall interference effects and supporting the assumption of negligible residual bias. The wind tunnel setup and scale model are shown in Fig. 5.

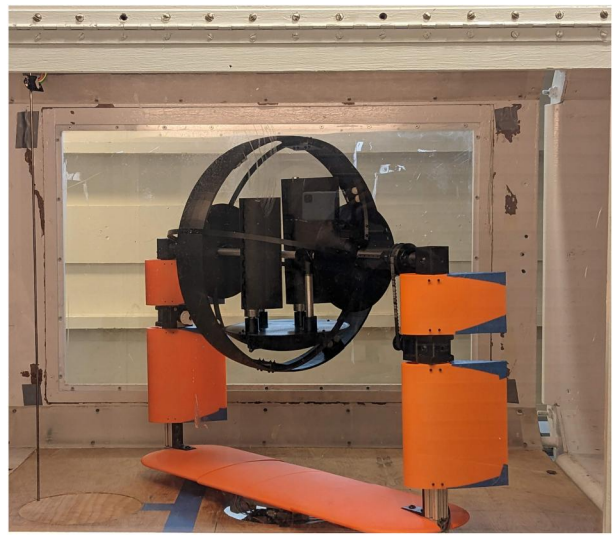

Fig. 5 The 40% scale model of the SSailOR on the test rig inside the wind tunnel [17]

To isolate the contributions of the various components of the SSailOR, in addition to identifying the effects of the mounting apparatus, it was essential to test a series of configurations. In particular, four configurations were considered:

- **Configuration 1:** Test of the mounting structure alone, for all tested yaw angles. This test provided base lift, drag, and yaw moment numbers to be subtracted from the measured values in subsequent tests.
- **Configuration 2:** Test of the complete structure of the SSailOR (Fig. 5, including the spherical cage (consisting of traction hoop and connecting ribs), the hull, and the jib and main sails. This test provided forces and moments acting on the combined cage (spinning and stationary), hull, and sails.
- **Configuration 3**: Test of only the traction hoop with connecting ribs, or just the spherical cage (i.e., no sails and no hull). This test is to isolate the effect of the spherical cage on windage drag and airflow when both stationary and rotating.
- **Configuration 4**: Test of configuration 2, with the SSailOR lowered on its mounting frame. This test confirmed that the aerodynamic forces and moments were only negligibly impacted by ground effects.
- **Configuration 5**: Test of only the hull, jib, and main sails (i.e., no traction hoop or connecting ribs, as shown in Fig. 6). This test serves to isolate the lift and drag of the inner assembly without the contribution of the spherical cage and to see if airflow is stifled by the outer apparatus.

Beyond the obvious need to subtract forces and moments from the mounting frame from the total forces and rule out the criticality of ground effects, the partitioning of testing into the above configurations served two important purposes. First, because the dynamic model characterizes the cage, hull, and sails independently, the isolation of force and moment contributions provides a sufficiently rich dataset for parameter identification. Secondly, because of the plastic structure of the wind tunnel model and its scale factor, some sacrifices needed to be made to the cage and the traction hoop. In particular, the traction hoop in the wind tunnel is one solid unit (rather than a series of tori), and the ribs comprising the cage are thicker. Isolation of force and moment contributions from the cage provided us with the ability to properly scale the drag contributions of the cage and traction hoop to reflect the full-scale design.

Within each of the aforementioned configurations, multiple conditions were varied to yield a sufficiently rich set of data for parameter identification. These conditions included varying dynamic pressure, spherical cage rotation (stationary and rotating, and only applicable for configurations 2 and 3, which involved the

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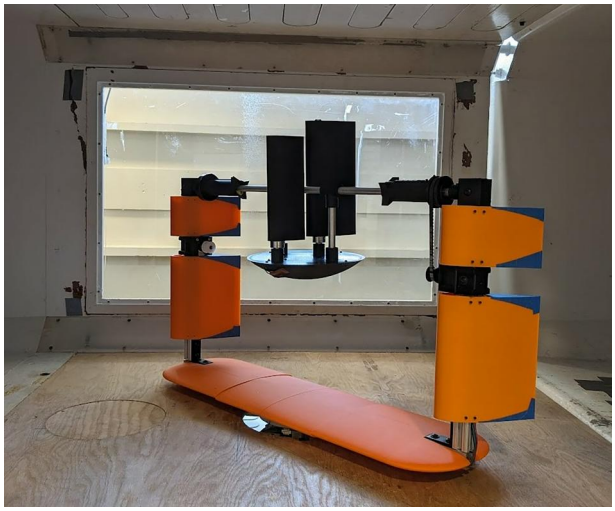

Fig. 6 The 40% scale model of the SSailOR with only the hull and sails for wind tunnel testing [17]

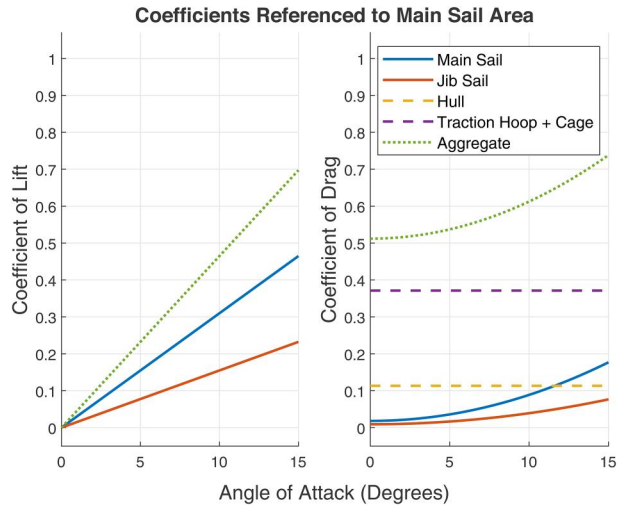

Fig. 7 Coefficients of lift and drag over a range of angles of attack

cage), varying sail angles with respect to wind direction, and varying yaw angle (which corresponds to variation in the apparent wind angle within the dynamic model). The full test matrix for each configuration can be found in Ref. [17].

4.2 Wind Tunnel Results. After compiling and processing the results of the wind tunnel testing, least squares parameter identification was used to characterize the sail's lift slopes, parasitic drag coefficients, Oswald efficiency factors, and the hull drag coefficient.

To account for potential wall interference effects caused by the wind tunnel confinement, blockage corrections were applied to the measured dynamic pressure. The blockage correction was applied using the relation introduced by Barlow et al. [18,19], which adjusts the dynamic pressure as follows:

$$q_c = q_u\left(1 + \frac{1}{2}\frac{A_f}{A_t}\right) \tag{25}$$

where q_u is the uncorrected dynamic pressure, q_c is the corrected dynamic pressure, and $\frac{A_f}{A_t}$ is the fractional blockage. The corrected dynamic pressure was used to calculate the lift and drag coefficients, and the updated values are reported in Table 6.

Because the wind tunnel model required that the traction hoop be solid and the cage be thicker than what is expected in the full-scale prototype, it was necessary to account for these geometric differences when estimating the aerodynamic coefficients. To do so, the lift slopes were calculated using both Configuration 2 and Configuration 5. A weighted average was used to determine the final lift slope, accounting for the difference in traction hoop and ribs for the wind tunnel model and the full-scale prototype. Meanwhile, the drag coefficients of the traction hoop and ribs were obtained from the literature [15].

Furthermore, the lift and drag coefficients for each rover component are plotted, along with the aggregate coefficients, in Fig. 7, all referenced to the main sail area to facilitate a direct comparison of the contribution of each component to the aggregate values.

Table 6 Values acquired from wind-tunnel tunnel experiments

Variable	Value	Units
k_1	1.775	rad^{-1}
$C_{D,0}$	0.0182	—
e	0.210	—
$C_{D,\text{hull}}$	0.0189	—

5 Dynamic Simulation Results

Leveraging the dynamic model introduced in Sec. 2 and the control strategy discussed in Sec. 3, simulations were conducted to characterize the SSailOR's performance. Prior to assessing the SSailOR's ability to track an upwind path, the relation between steady-state rover speed and apparent wind speed and direction was characterized. This characterization was performed for a single plant design, as described in Table 7. Following this characterization, the ability of the system to tack and track an upwind path was evaluated. The heel angle threshold, $\beta_{\max}$, is set to 9 degrees, which is below the limit of 15 degrees, wherein the traction hoop loses contact with the ground. This provides a buffer, which allows the SSailOR to continue to operate within its physical heel limits even in the presence of large disturbances and transients due to events such as heading changes. Rate limits were placed on the heading setpoint and sail angles. The heading setpoint was given a rate limit of ± 20 deg/s (0.349 rad/s) so that the SSailOR would not take instantaneous turns and undergo a centripetal force resulting in excessive heel. The sail angles were given a rate limit to reflect actuator speeds. Because the mast is assumed at the quarter chord, loads on the servos are expected to be low. Based on the common 10 kg × cm servos available, a rate limit of ± 352.9 deg/s (6.16 rad/s) was applied to the sail angles. Finally, the tradeoff between maximizing steady-state velocity and minimizing rise time (the time needed for the rover to reach 90% of its steady-state velocity) when tracking a straight-line path was analyzed across a range of rover designs via a Pareto analysis.

5.1 Straight Line Tracking and Velocity Polar. Before assessing the ability of the SSailOR to change directions via tacking, it was first necessary to validate the closed-loop system's ability to track a constant heading. To this end, simulations were conducted wherein the rover started stationary, and a constant heading angle setpoint (corresponding to a straight-line path) was tracked at wind speeds ranging from 4 m/s to 10 m/s. This simple initial exercise was performed with the wind directly perpendicular to the rover ($\psi_{\text{rel}} = \frac{\pi}{2}$). The achieved speed of the SSailOR over the course of these simulations is plotted in Fig. 8, with the corresponding jib and main sail angle and the angle of attack trajectories shown in Figs. 9 and 10, and the heading error trajectories shown in Fig. 11. Two key observations can be gleaned

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Table 7 Plant and controller parameters used in dynamic simulations

Variable	Value	Units
R	0.5	m
τ	0.001	m
n	16	—
τ_{support}	0.0075	m
m_{hull}	4	kg
m_{cage}	1	kg
H	0.191	m
b_{jib}	0.6	m
AR_{jib}	3.429	—
b_{main}	0.6	m
AR_{main}	2	—
S_{jib}	0.105	m^2
S_{main}	0.18	m^2
$k_{1,\text{jib}}$	1.775	rad^{-1}
$k_{1,\text{main}}$	1.775	rad^{-1}
e	0.2104	—
$C_{\text{D},0_{\text{jib}}}$	0.0182	—
$C_{\text{D},0_{\text{main}}}$	0.0182	—
$C_{\text{D,hull}}$	0.0189	—
ρ	1.225	$\text{kg} \cdot \text{m}^{-3}$
g	9.81	$m \cdot s^{-2}$
C_{rr}	0.01	—
ℓ_{jib}	0.3	m
ℓ_{main}	0.134	m
$d_{\text{qc}_{\text{jib}}}$	0	m
$d_{\text{qc}_{\text{main}}}$	0	m
$\beta_{\max}$	0.1571 (9)	rad (deg)
w	1	—
K_{P1}	20	$\text{rad} \cdot \text{rad}^{-1}$
K_I	5	$\text{rad} \cdot \text{rad}^{-1} \cdot \text{s}^{-1}$
K_D	20	$\text{rad} \cdot \text{rad}^{-1}\ \text{s}$
N	100	—
G_2	1	$\text{s} \cdot \text{m}^{-1}$
$\alpha_{\max,\text{jib}}$	0.262 (15)	rad (deg)
$\alpha_{\max,\text{main}}$	0.262 (15)	rad (deg)
θ	0.3 (17.189)	rad (deg)

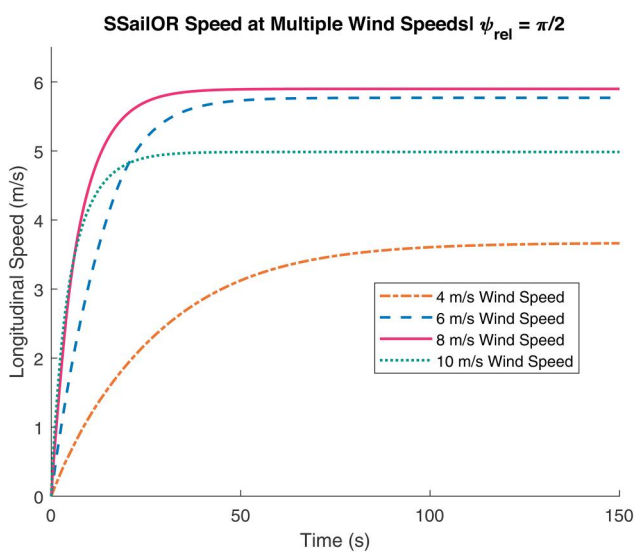

Fig. 8 Longitudinal speeds over time at 4, 6, 8, and 10 m/s

from these simulation results: (i) A constant heading can be accurately tracked using the proposed control architecture, and (ii) there arises a performance tradeoff due to the heel controller, where too much force generated by a sail requires very suboptimal angles of attack to stay below a certain heel angle. This is the reason that the SSailOR actually attains higher steady-state speeds in 6 and 8 m/s wind than it does in 10 m/s wind.

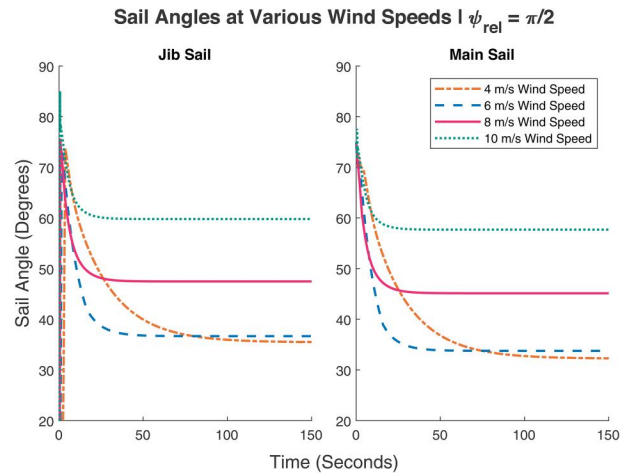

Fig. 9 SSailOR jib and main sail angles over time at 4, 6, 8, and 10 m/s

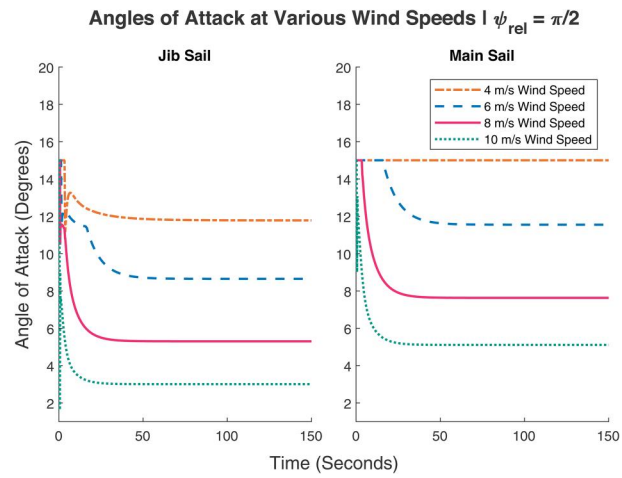

Fig. 10 SSailOR jib and main angles of attack over time at 4, 6, 8, and 10 m/s

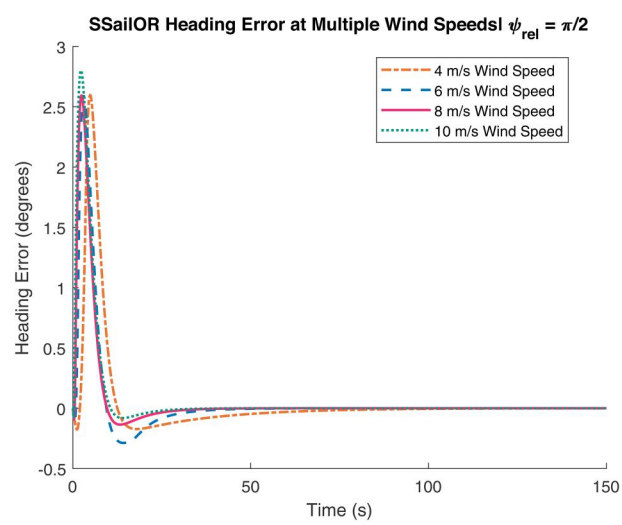

Fig. 11 Heading error over time at 4, 6, 8, and 10 m/s

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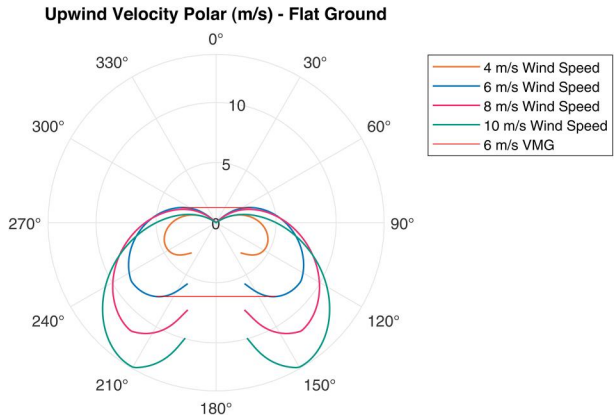

Fig. 12 Velocity polar showing steady-state rover speeds for incident wind angles at 4, 6, 8, and 10 m/s. Example VMG ("velocity made good") shown for the 6 m/s case.

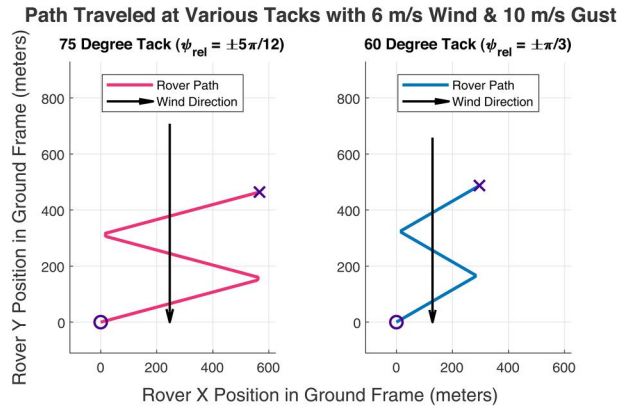

Fig. 13 Upwind path traveled by the SSailOR with a 6 m/s wind speed and 10 s gust of 10 m/s

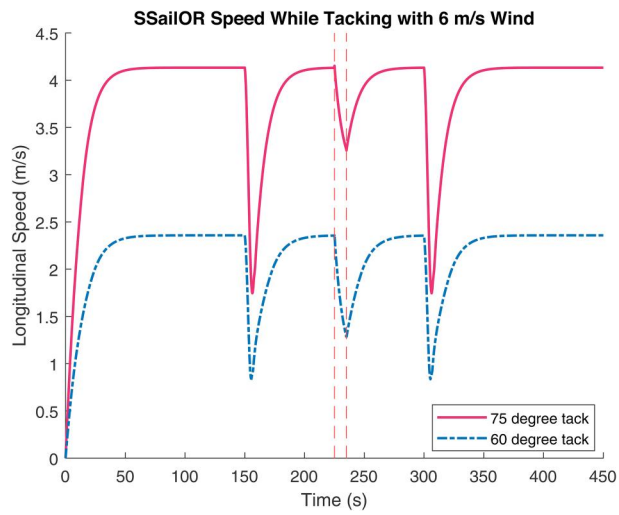

Fig. 14 SSailOR speed over time in tacking simulations

After validating the rover's ability to track a constant heading, peak rover speed was characterized for a range of wind speeds and directions. This characterization was done through the construction of a velocity polar plot, which is an established plot in sailboat racing literature that is used to characterize a vessel's steady-state performance [13]. The velocity polar plot provides the steady-state speed reached at every true wind angle relative to the SSailOR, for a variety of wind speeds, providing insights into the best-case performance of a given design. A velocity polar of the device described in Table 7 is shown in Fig. 12, considering wind speeds ranging from 4 to 10 m/s. Note that the velocity polar plot does not contain results for conditions where the rover is operating nearly downwind, due to stall being unaccounted for in the dynamic model (which renders the dynamic model inaccurate for large angles of attack that are inevitable in direct-downwind operation). This occurred at wind angles of 142 deg, 156 deg, 163 deg, and 166 deg for wind speeds of 4, 6, 8, and 10 m/s, respectively. However, these extreme downwind scenarios are also not particularly significant for this design, due to the "Velocity Made Good" or VMG, which is shown in red for a wind speed of 6 m/s in Fig. 12. The VMG represents the maximum *component* of velocity achievable in a particular direction. For directions where the VMG exceeds the velocity indicated by the polar plot (including extreme downwind conditions), the SSailOR should tack (if upwind) or jibe (if downwind) rather than pursuing a straight-line path.

5.2 Upwind Path Tracking Via Tacking With Gusts. After demonstrating the SSailOR's ability to track constant headings, the ability of the SSailOR to track a nonconstant, upwind path with gusts of wind as disturbances was demonstrated. Simulation results are presented wherein the SSailOR traversed two upwind paths at a constant wind speed. The first of these paths required the SSailOR to complete two 75 deg tacks, while the second required the SSailOR to complete two 60 deg tacks (at a wind speed of 6 m/s in both cases). In both tacking scenarios, the rover experiences a sudden gust of 10 m/s wind that occurs 225 s into the simulation and lasts 10 s. The upwind paths traversed by the SSailOR in these simulations are shown in Fig. 13, with the corresponding speed, heel angle, heading error, sail angles, and angles of attack shown in Figs. 14–18.

As is demonstrated in Fig. 13, the SSailOR follows a smooth path when tacking. At the same time, the heel angle of the rover (as depicted in Fig. 3) closely tracks $|\beta_{\max}|$, indicating that the heel angle controller functions as intended. As shown in Fig. 14, while higher steady-state speeds are achieved at more aggressive apparent wind angles, the speed reduction during tacking is more pronounced at these angles. This is due to the greater extent of velocity redirection required to complete larger tacks. In addition, the figures show the robustness of the SSailOR to sudden and formidable gusts of wind. In particular, these gusts are less disruptive and impactful than the act of tacking itself.

5.3 Transient/Steady-State Tradeoff Analysis. In this subsection, we assess the impact of critical design variables on the SSailOR's steady-state speed (V_{ss}) and its ability to get up to speed quickly, as reflected through its rise time (t_r). Although several design parameters of the SSailOR impact V_{ss} and t_r for a given wind speed, the most impactful parameters based on initial observation are the total mass, m, and the aspect ratio of the sails, AR, under the assumption of constant span. The assumption of a constant span is made for two reasons: (i) Keeping the sails at the largest span possible within the spherical cage without interfering with airflow onto the wing is intuitive, as the wing will have one dimension of its planform area as large as possible for performance, and the other dimension can be modified to lean toward efficiency or more force generation. (ii) Due to the numerous factors that could affect the performance of the SSailOR—and a desire to keep simulation times and complexity reasonable—there is a need to bound one of the dimensions making up the sail's planform area. The span is mainly limited by the spherical cage's dimensions, whereas the chord dimensions of both the jib and main sails have to consider the spherical cage, their quarter chord distance from the center (and their impact on heading control), and their interference with one another, both physically and aerodynamically. Other factors, including but not limited to the wind direction (ψ_{rel}), are challenging to incorporate into a brief tradeoff analysis and are deserving of future experiment in their own right.

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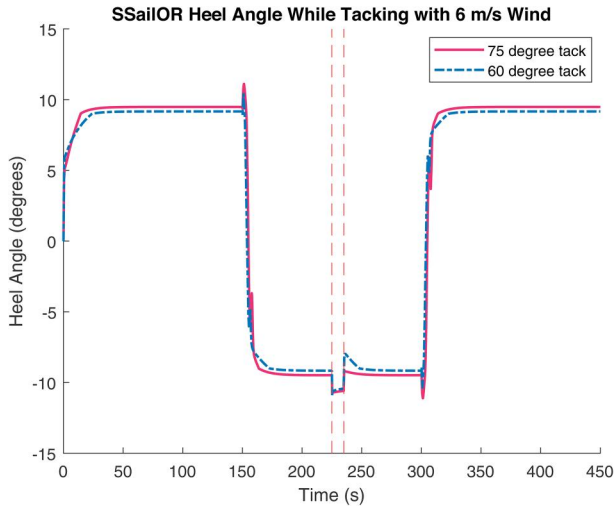

Fig. 15 Heel angle over time in tacking simulations

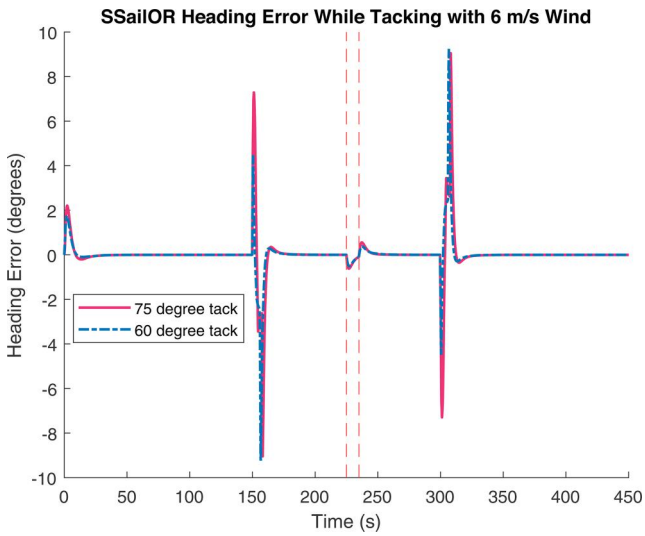

Fig. 16 Heading error over time in tacking simulations

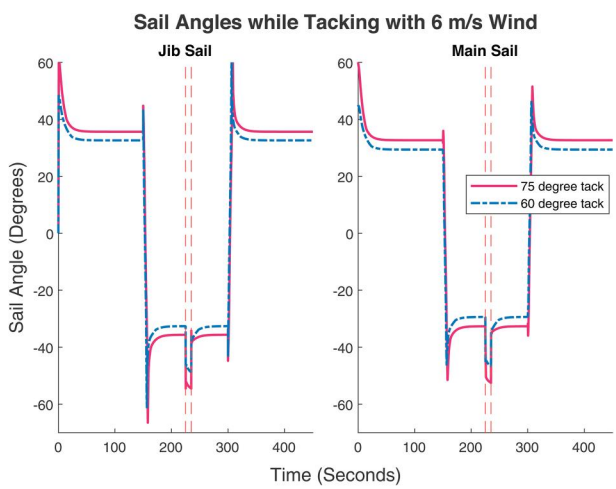

Fig. 17 Jib and main sail angles over time in tacking simulations

Intuitively, increasing the mass of a system will increase rise time. That said, as a result of a constraint on the maximum heel angle, increasing mass will also increase steady-state speed. As rover mass increases, so do the lateral forces needed to drive the rover to heel. If

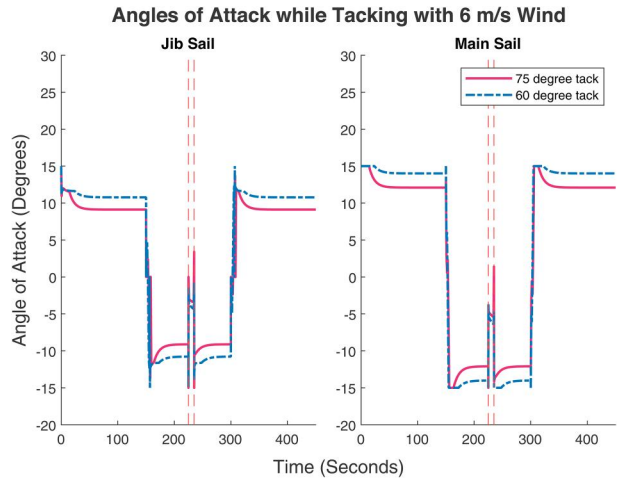

Fig. 18 Jib and main sail angles of attack over time in tacking simulations

the rover mass is small, the angle of attack must be curtailed to keep the heel angle at or below $|\beta_{\max}|$. At higher masses, the angle of attack that maximizes longitudinal force (and thus steady-state speed) can be tracked more closely without violating the bounds on allowable heel angle. That said, the relationship between aspect ratio, rise time, and steady-state speed is much less intuitive. Under the assumption of a constant span, decreasing aspect ratio leads to both an increase in sail area and a decrease in aerodynamic efficiency. As a result, decreasing the aspect ratio results in faster rise times but lower steady-state speeds.

To quantify the coupling between mass and aspect ratio with the competing objectives of rise time and steady-state speed, a Pareto analysis was performed at multiple wind speeds, considering the following competing objective functions (where lower values are favorable for each):

$$\begin{aligned} J_{\text{Transient}} &= t_r(m, AR) \\ J_{\text{SteadyState}} &= \frac{1}{V_{\text{ss}}(m, AR)} \end{aligned} \tag{26}$$

To generate the simulation results needed to carry out this optimization, a sweep of simulations was conducted at $\psi_{\text{rel}} = \frac{\pi}{2}$ and multiple wind speeds between 4 and 10 m/s, where the mass was varied from 3 kg to 20 kg, and the aspect ratio was changed via a chord length scale factor. The span of both the jib and main sail was set to 0.600 meters, and the chord lengths of the jib and main sail in the Sec. 5.1 and 5.2 simulations were multiplied by a length scaling factor, which varied from 0.4 to 1.1. These values were chosen based on the minimal performance required to achieve any motion at 10 m/s, as well as geometric constraints to still fit within the spherical cage. Using these simulation results, the quantities described in Eq. (26) were computed over a range of wind speeds. The results of this analysis are shown in Fig. 19. In this figure, the size of each point corresponds to the mass of each design, while the color of each point corresponds to its chord length scale factor. Note that at the minimum scaling factor, the aspect ratios of the jib and main sail were 8.57 and 5, respectively. At the maximum scaling factor, the aspect ratios of the jib and main sail were 3.11 and 1.82, respectively.

Several interesting insights can be gleaned from this analysis. First off, at 4 m/s wind speed, fewer data points are visible because many of the tested designs performed poorly and fell outside the plotted range, making them nonviable. Looking again at the 4 m/s wind speed, the Pareto front is very small and narrow, favoring light crafts with smaller aspect ratios (i.e., larger chord length scaling factors). At this speed, a large amount of force and light weight is preferred for overcoming rolling resistance, and since the heel angles at these low wind speeds are small, large sails are not a

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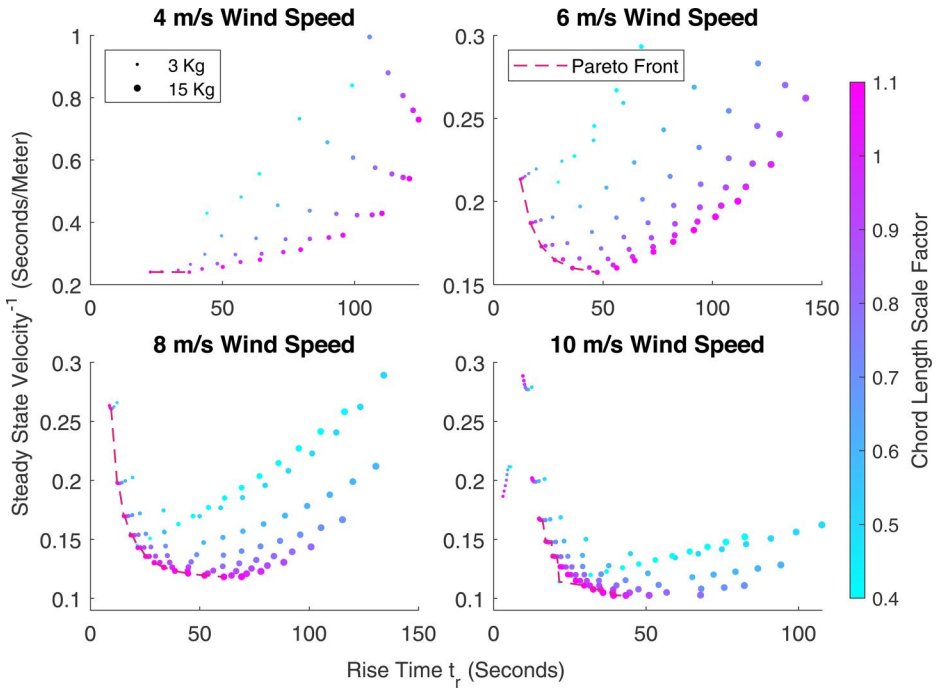

Fig. 19 Rise time and inverse steady-state velocity across varying masses and chord length scaling factors at 4, 6, 8, and 10 m/s, with $\psi_{rel} = \frac{\pi}{2}$. Dashed lines show the Pareto fronts, consisting of all nondominated points in the lower-left region of each plot.

detriment to performance. At low wind speeds, the heel angle constraint is often inactive, even when considering low-mass designs. Given that reducing mass reduces rise time, a lightweight design is preferable when designing a system to operate in low wind speeds. The opposite is true when designing a device to operate in high wind speeds. In this environment, a large mass is required so the sails can operate at angles of attack near α_{opt} without curtailing the angle of attack to satisfy the heel angle constraint. Generally speaking, for every wind speed, powerful low-aspect ratio sails were preferred—likely to overcome rolling resistance and windage drag. It is noteworthy, however, that reduced windage drag is likely to result in a preference for higher aspect ratio sails in order to maximize steady-state speed. This is because higher aspect ratios lead to higher lift-to-drag ratios, in spite of having less area (when the chord length is reduced to increase the aspect ratio).

6 Conclusions and Future Work

Research in remote regions with hostile climates is difficult and expensive. Research missions falling within this category include Arctic/Antarctic climate change characterization and desert exploration. To alleviate the burden on researchers, autonomous rovers have received significant attention. However, current solutions are unable to carry out persistent unmanned missions due to design flaws, ranging from tipping under high winds to the inability to make upwind progress. In this paper, we presented a control-oriented dynamic model, control system design, wind tunnel test results, and several steady-state and dynamic analyses for the SSailOR—the first wind-powered spherical rover that is capable of achieving net upwind motion. The dynamic model makes it possible to characterize both the transient and steady-state response of the SSailOR under closed-loop control, whereas the presented control architecture contains mechanisms for optimally trimming the sails, tracking a target heading, and mitigating heel angle. The simulation-based analysis results characterize the SSailOR's steady-state and dynamic behavior in addition to providing insight into critical design trades between mobility (transient response) and steady-state performance (along with the design considerations that are likely to favor one attribute of performance over another). The results of the simulation-based analysis are strengthened by validation through wind-tunnel validation, which helped refine the model and elevate the design's feasibility.

Future work will focus on enhancing the fidelity of the dynamic model (including variable wind shear profiles/gradients), considering different topographies in simulation (such as uphill and downhill movement and the inclusion of obstacles), detailed ground interaction in various terrain, such as soft soil and snow, and experimental demonstration of upwind capabilities in the North Carolina Outer Banks. As the fidelity of the dynamic model is expanded, the controller's performance will be monitored to ensure that the current architecture remains effective. If performance begins to suffer, particularly in terms of constraint satisfaction, other control architectures (such as model predictive control (MPC)) could be considered to meet system objectives. Once thoroughly tested on the Outer Banks, long-term future deployment scenarios will target missions in the Arctic tundra and Antarctic to characterize the ability of the SSailOR to handle persistent missions in more extreme conditions. Additional planned future work will include more detailed trade studies for optimizing the performance of the SSailOR—including extension of the aforementioned Pareto analysis to consider different wind directions. This will enable us to determine how the optimal values of key design parameters (like mass and aspect ratio, among others) change as functions of the wind direction. Future design efforts will focus on leveraging these Pareto analysis results to optimize a design based on the wind statistics of a target mission domain, in addition to considering very practical matters such as the structural optimization of the traction hoop and cage to meet structural loads with the thinnest (and therefore least aerodynamically impactful) hoops possible. Finally, future research will focus on the use of the control notion of reachability to formally quantify and manage the tradeoff between mobility and steady-state behavior in the SSailOR.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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